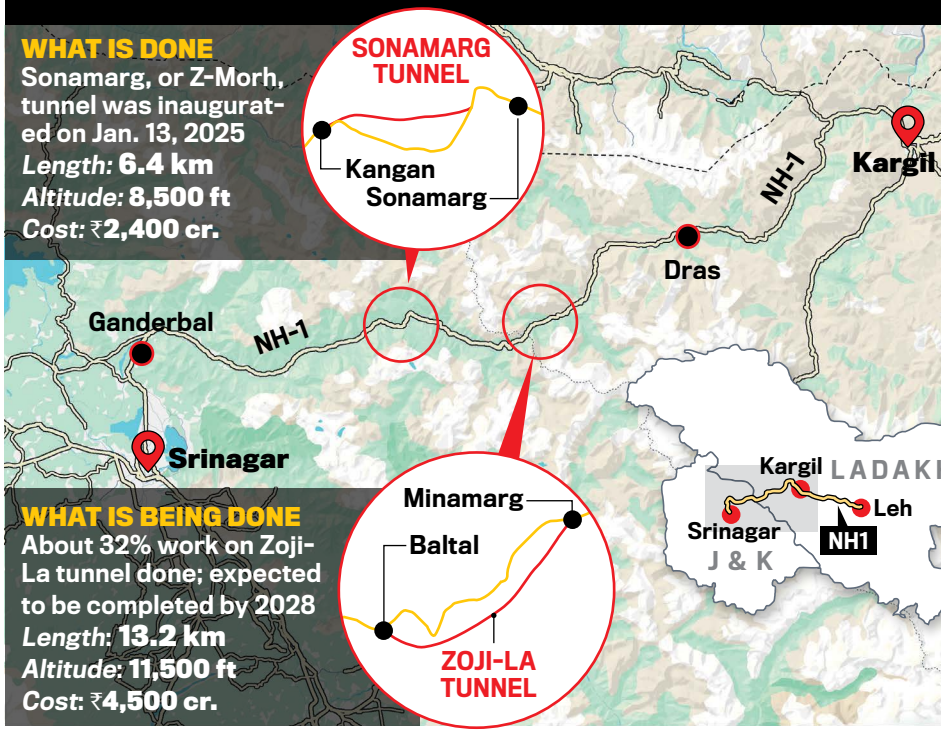
THE HIMALAYAN FEAT

The opening of Sonamarg tunnel in central Kashmir marks a major milestone in the eventual all-weather road connectivity to Ladakh



Graphic by TANMOY CHAKRABORTY

THE MANY CHALLENGES

- **Freezing conditions in the region, nosediving to minus 30 degrees in winters**
- **Excavation through daunting rocks and a 60-degree slope surface that had to be stabilised**
- **Frequent landslides, like one in Feb. 2023 that buried the work 80m under the soil**
- **Security concerns dissuaded workers; almost 60% were unskilled locals**

tourist attraction in winters, and now there is finally an enabling factor. "The tunnel will ensure connectivity, helping people and businesses thrive even in winter," he adds.

Constructed at a cost of Rs 2,400 crore, the Sonamarg tunnel's journey from concept to completion wasn't without obstacles, recalls Vijay K. Pandey, General Manager–Projects, National Highways and Infrastructure Development Corporation Limited. Initially conceived in 2012, multiple hurdles delayed things, and work finally began in June 2020. A terror attack on its Gagangir campsite in October 2024 claimed seven lives, while a massive landslide disrupted work in 2023. Regardless, the project pressed forward, driven by the determination of 1,500 workers and 300 engineers. "Once the machines are switched off, restarting them becomes extremely challenging due to the harsh cold. To avoid this, workers had to maintain operations continuously, day and night," says Pandey.

The tunnel's advanced design includes a 10-metre-wide bidirectional main passage, an escape tunnel with emergency exits every 250 metres, 350 surveillance cameras and a modern ventilation mechanism—a robust infrastructure meant to withstand even extreme conditions. Including its 5.5-km approach roads, the tunnel is expected to reduce travel time on this stretch from over two hours to just 15 minutes.

The ultimate goal remains the completion of the Zoji-La tunnel, which bypasses the Zoji-La pass, one of India's highest motorable passes and presently the gateway to Ladakh. It, too, gets cut off every winter. So far, 79.5 per cent of the excavation work has been completed while overall 32 per cent work is done. "Here, the poor geological condition of rocks with horizontal and vertical joints is a different ball game altogether. Also, there are huge water reservoirs inside the mountains, which need effective diversion," says Yousef Eshaghpour, an Iran-based design engineer overlooking the project. While officially it has a September 2027 deadline, Eshaghpour predicts the Zoji-La tunnel won't be ready before December 2028.

The two tunnels are vital not just for regional connectivity but also for countering China's infrastructure buildup along the LAC after the 2020 Galwan standoff. "These tunnels will significantly bolster the Indian armed forces' response across the LAC, LoC and other key points," says Lt Gen. D.S. Hooda (retired). "The resources required for the long summer logistical exercise of stocking weaponry, fuel and eatables for winter months can be repurposed for other vital aspects to protect our sovereign interests."

Keen observers view Modi's presence at the tunnel's inauguration, just three months after the deadly terror attack, as a powerful message—that the region's integration and development will proceed undeterred. On January 6, the PM had virtually inaugurated the Jammu Railways Division. Next in line is the New Delhi-Srinagar train service, which promises to further connect the Valley with the rest of India. ■